

claude-windows / da34752f-c664-4262-9602-f5d4cfcc6db4

Metadata

```
- Source: `claude-windows`  
- Kind: `claude`  
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\da34752f-c664-4262-9602-f5d4cfcc6db4\subagents\workflows\wf_774b13f4-334\agent-a7b5440f4f57f3371.jsonl`  
- SHA-256: `30ae90d495d008694526b0a698b3372354efd9e806638042d5f15627608e1e78`  
- Source modified: `2026-06-12T14:39:30+00:00`  
- Imported at: `2026-07-05T16:48:22+00:00`  
- project: `wf_774b13f4-334`  
- session_id: `da34752f-c664-4262-9602-f5d4cfcc6db4`
```

Transcript

1. user (2026-06-12T14:36:57.598Z)

Repo: C:\Users\avidu\Projects\badminton-highlight-indexer (branch docs/pmf-field-signals).

Review these files (read them in full):

- docs/PMF_FIELD_SIGNALS.md (NEW - hypotheses H1-H7, GREEN/AMBER/RED scoring anchors, pre-registered decision rules PROCEED/PIVOT A-C/FALSIFIED, rollup mechanics, guardrails)
- docs/FIELD_VISIT_ANSWER_SHEET.md (NEW - the per-visit capture form a non-technical field associate fills by hand)
- docs/FIELD_RESEARCH_ASSOCIATE_FLYER.md (EXISTING, lightly edited - the recruit hand-out with the 7-question script and ground rules)

Grounding docs (also read): docs/PMF_MARKET_RESEARCH.md (esp. SS6 risks and pricing anchors), docs/NEXT_STEPS.md item 5, docs/COMMERCIALIZATION.md C13 row.

Context: the owner ratified that PMF validation happens by physically visiting Bangalore badminton academies (10-15 visits by a paid non-technical field associate) and reading signals. The kit's job: make 15 visits produce a pre-registered verdict (proceed/pivot/falsified) rather than motivated reasoning. Product = KhelSutra (rally indexing + highlights from static-camera footage; wedge = messy multi-court BYO footage; serving COGS ~\$0.30-0.60/hr via Gemini).

A critic claims the following flaw in the field-validation kit. Adversarially verify by reading the actual files - try to REFUTE it. Default to refuted if the flaw is pedantic, out of scope for a 10-15-visit directional read, or already mitigated elsewhere in the kit.

CLAIM [low] The opening line pitches the solution before the problem questions

The flyer's opening ("an AI tool... takes normal practice video and automatically cuts it into rallies and highlights") describes the product before Q1?Q3 are asked. Mom Test discipline is to talk about their life before your idea: once the respondent knows what you hope is true, problem questions (do you have dead footage? who'd watch clips?) collect courtesy-shaped answers ? particularly under Indian business politeness norms, where a direct "no" to a guest is rare. The required-negative-quote quota partially mitigates, but the GREEN tallies still count primed positives. Honesty about purpose doesn't require revealing the wedge.

Proposed fix: Reorder: open with "helping an engineer study how academies record and use video ? not selling anything", run Q1?Q2 (pure past-behavior), THEN reveal the rally/highlight concept before Q3?Q7 (which inherently need it). Note on the sheet where the reveal happened so the scorer knows which answers were pre- vs post-pitch.

2. user (2026-06-12T14:37:05.002Z)

docs/pmf-field-signals

M docs/FIELD_RESEARCH_ASSOCIATE_FLYER.md

M docs/NEXT_STEPS.md

?? arial.ttf

?? artifacts/

?? docs/FIELD_VISIT_ANSWER_SHEET.md

?? docs/PMF_FIELD_SIGNALS.md

?? scratch/HIGH_RISK_REMEDIATION_PLAN.md

```
?? scratch/PR1_BODY.md
?? scratch/PR1_HANDOFF.md
?? scratch/PR2_HANDOFF.md
?? scratch/PR2_deferred_hardening_status.patch
?? scratch/PR88_FIX.md
```

3. user (2026-06-12T14:37:09.573Z)

```
1      # C13 Field Signals ? what academy visits must tell us (and when to change course)
2
3      **Status:** ACTIVE ? the decision layer for the C13 beachhead validation
4      (owner-ratified direction, 2026-06-12: *PMF validation is handled by physically
5      visiting academies and reading signals on the ground*).
6      **Companions:** [`FIELD_RESEARCH_ASSOCIATE_FLYER.md`](FIELD_RESEARCH_ASSOCIATE_FLYER.md)
7      (the recruit hand-out: script + ground rules) .
8      [`FIELD_VISIT_ANSWER_SHEET.md`](FIELD_VISIT_ANSWER_SHEET.md) (the per-visit capture
form) .
9      [`PMF_MARKET_RESEARCH.md`](PMF_MARKET_RESEARCH.md) (the hypotheses under test, esp. §6).
10
11     **Why this doc exists:** the flyer collects answers; this doc fixes ? **before the
12     first visit** ? what those answers *mean* and what totals trigger which decision.
13     Pre-registering the thresholds is the honesty mechanism: it stops us from reading
14     15 ambiguous conversations as whatever we hoped to hear.
15
16     ---
17
18     ## 1. The hypotheses under test (each maps to script questions)
19
20     | ID | Hypothesis (from PMF research) | Falsified looks like | Script Q |
21     |---|---|---|---|
22     | H1 | **Recording behavior exists** ? academies already produce fixed-angle / CCTV /
multi-court footage without us | nobody records; "we'd have to start for you" | Q1 |
23     | H2 | **Dead-footage pain** ? recorded footage piles up unwatched (the indexing wedge)
| footage is small, or already gets watched | Q2 ? |
24     | H3 | **An audience exists** ? coach, player, or parent would actually consume rally
clips/reels | shrug; "who has time to watch clips" | Q3 |
25     | H4 | **Whitespace** ? incumbents (Machaxi, VISIST, Game Theory, Stupa, BadPro+) leave
a gap we fit | "we use X, it's fine, fully solved" | Q4 |
26     | H5 | **Willingness to pay per match** at a price that clears unit economics | "should
be free" / <?100 everywhere | Q5 ? |
27     | H6 | **Multi-court chaos is a felt problem** (our moat: indexing messy BYO multi-court
footage) | multi-court video already useful to them | Q6 |
28     | H7 | **Behavioral commitment** ? they'll let us record on their court (Phase-2 door) |
polite no across the board | Q7 |
29
30     Unit-economics floor behind H5: serving COGS on the Gemini path is ~$0.30?0.60 per
31     hour of video (? ?25?50/hr; 2026-06-10 quality table). A 2?3 hr session costs
32     roughly ?75?150 to process, so **?300/match has real margin, ?100/match is
33     break-even-ish at retail** ? that's why the script's anchor ladder is ?100/300/500.
34
35     ## 2. Per-visit scoring ? GREEN / AMBER / RED anchors
36
37     Score each hypothesis on the answer sheet **from what they said and showed, not
38     what you inferred**. When between two grades, take the lower one.
39
40     | | GREEN (strong confirm) | AMBER (weak/mixed) | RED (disconfirm) |
41     |---|---|---|---|
42     | **H1 record** | shows or names actual setup (tripod/CCTV/phone rig) used ?monthly |
records occasionally (tournaments only) | doesn't record; no plans |
43     | **H2 dead footage** | points at real unwatched volume (?~5 hrs, "CCTV runs all day",
tournament backlog) | some footage, partially watched | no backlog exists |
44     | **H3 audience** | names WHO would watch and WHEN (e.g. "parents ask for clips", "I
review with students Mondays") | "could be useful, maybe" | no consumer identified |
45     | **H4 whitespace** | no tool today, or uses one and volunteers a gap we cover (multi-
court, BYO footage, price) | uses a competitor, mildly satisfied | competitor solves it,
```

satisfied, sticky |

46 | **H5 WTP** | volunteers ??300/match unprompted, or proposes a budget framing of their own (per-student/per-month) | lands at ~?100 with hesitation | refuses to name a number / "free" |

47 | **H6 multi-court** | complains about it in their own words before/without prompting | agrees when asked | multi-court footage already works for them |

48 | **H7 Phase-2** | yes + gives a name/number/slot | "maybe, come back" | no |

49

50 **Weighting rule (behavior > words):** H7-GREEN and H2-GREEN are *behavioral* signals and count double in your gut-read; H3/H5 answers are *talk* until a Phase-2 session or a real payment corroborates them. A visit with H5-GREEN but H7-RED is a polite conversation, not a customer.

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55 **Disconfirming-evidence quota:** every visit sheet must record at least one verbatim *negative* quote (the thing they liked least / wouldn't pay for). If a sheet has only positives, the visit gets re-done or discounted ? per the flyer's ground rule 4, a reasoned "no" is worth more than five vague yeses.

56

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59

60 **## 3. Pre-registered decision rules (n = 10?15 academies)**

61

62 Counts, not percentages ? at this sample size we're reading direction, not estimating proportions. Evaluate after **every 5 completed sheets**; final call at the agreed n. "GREEN" below means GREEN on that hypothesis row.

63

64

65

66 | Verdict | Trigger (pre-registered) | What we then do |

67 |---|---|---|

68 | **PROCEED ? beachhead validated** | ?6 H2-GREEN **and** ?4 H5 at ?300+ **and** ?3 H7-GREEN | Run the camera pilot at the ?3 warmest academies; those contacts seed the UI-alpha Wave-1 invite list (UI_ALPHA_EXECUTION_PLAN Phase R); start the paid-pilot conversation with the single warmest |

69 | **PIVOT A ? "index the backlog" ? "record + index" service** | H2 mostly RED (<3 GREEN) but H3+H5 healthy on *fresh* recordings | The pain isn't the archive, it's that nothing useful gets made at all ? lead with the Phase-2 recording offer; camera pilot becomes the product wedge, not the demo |

70 | **PIVOT B ? partner/API motion** | H4 RED at ?5 academies (incumbents entrenched and liked) | Don't fight installed hardware head-on; explore the engine/API route through whoever owns the academy relationship (the PB Vision x PodPlay template from the 2026-06-10 audit) |

71 | **PIVOT C ? payer is the parent, not the academy** | H5 RED from owners but H3 repeatedly names parents/players asking for clips | Re-test WTP as B2C per-student reels (?/reel to parents) with the academy as channel, not customer |

72 | **FALSIFIED ? beachhead assumption fails** | <3 academies record at all (H1) **and** 0 H7-GREEN | The "academies already record fixed-angle" premise (PMF §6 risk 1) is wrong for this market ? revisit beachheads #2/#3 (pickleball-US, badminton Asia/Denmark) before building further |

73

74 Mixed outcomes that match no row: write up the per-hypothesis tallies and decide explicitly with the owner ? but **the tallies above get computed first**, so the discussion starts from the pre-registered read, not from anecdotes.

75

76

77

78 **Competitor-intel side-channel (always-on):** any visit where a competitor is named ? fill the answer sheet's competitor block (what it does, what it misses, what they pay). ?3 consistent reports about one product = update

79

80 ``PMF_MARKET_RESEARCH.md`` §4 with field-verified rows (today's table is web-sourced).

81

82

83 **## 4. Mechanics**

84

85 - One [`FIELD_VISIT_ANSWER_SHEET.md``](FIELD_VISIT_ANSWER_SHEET.md) per academy, same day as the visit, in the associate's words + verbatim quotes.

86

87 - The owner (or Claude session) transcribes each sheet into one row of the rollout table below (keep it in this doc or a sheet next to it); tallies and the decision-rule read-out update per 5 visits.

88

89

90 - Demo asset if a conversation earns it: the real compiled reel (`output/MahadevpuraSingles_highlights.mp4``) on a phone ? *show, never promise* (flyer ground rule 3 still applies; the associate doesn't carry the demo, the owner does on follow-ups).

91

92

93

```

94
95     ### Rollup table (one row per completed visit)
96
97     | # | Academy | Area | Courts | Role spoken to | H1 | H2 | H3 | H4 | H5 (?) | H6 | H7 |
Gut | Phase-2? |
98     |---|---|---|---|---|---|---|---|---|---|---|---|
99     | 1 | | | | | | | | | | | |
100
101     ### Tally ? decision read-out (recompute per 5 visits)
102
103     | Metric | Count | Threshold in play |
104     |---|---|---|
105     | H2 GREEN (dead footage) | /n | ?6 ? PROCEED leg; <3 ? PIVOT A leg |
106     | H5 at ?300+ | /n | ?4 ? PROCEED leg |
107     | H7 GREEN (Phase-2 yes + contact) | /n | ?3 ? PROCEED leg; 0 ? FALSIFIED leg |
108     | H4 RED (incumbent satisfied) | /n | ?5 ? PIVOT B |
109     | H1 any recording | /n | <3 ? FALSIFIED leg |
110     | H5 RED + H3 names parents | /n | recurring ? PIVOT C |
111
112     ## 5. Guardrails that bind the field work
113
114     - **Phase 1 records nothing** ? notes only (flyer rule 7).
115     - **Phase 2 consent rules are hard:** written consent from the academy AND adult
116       players; **sessions with minors are never recorded for our library** (training-pool
117       guardrail, CLAUDE.md); the academy is told plainly what the footage is for.
118     - Answer sheets contain personal contact details ? they stay between the associate
119       and the owner; transcribe to the rollup without phone numbers.
120

```

4. user (2026-06-12T14:37:10.051Z)

```

1     # Field Visit Answer Sheet ? one per academy (fill the same day)
2
3     > Print one per planned visit. Write THEIR words, not yours. A clear "no, because?"
4     > is a *good* sheet. Companion script: `FIELD_RESEARCH_ASSOCIATE_FLYER.md`.
5     > Scoring anchors (G/A/R): `PMF_FIELD_SIGNALS.md` §2.
6
7     ## Visit basics
8
9     | | |
10    |---|---|
11    | Date / time | |
12    | Academy name | |
13    | Area (neighbourhood) | |
14    | # courts in the hall | |
15    | Spoke to (name + role) | |
16    | Contact (phone/WhatsApp) ? *only if offered* | |
17    | How long the conversation ran | |
18
19    ## The seven questions ? their answers (verbatim where possible)
20
21    **Q1 ? Do they record sessions/matches today? How (phone / tripod / CCTV / hired)?**
22
23    >
24
25    **Q2 ? ? Footage sitting around that nobody watches? Roughly how much, and from what?**
26
27    >
28
29    **Q3 ? Who would watch rally clips/highlights (coach / players / parents)? When?**
30
31    >
32
33    **Q4 ? What do they use today (Machaxi / VISIST / Game Theory / BadPro+ / none)?
34    What's good, what's missing, what does it cost them?**

```

```

35
36 >
37
38 **Q5 ? ? Would they pay per match for rally cuts + reel + report? What number did
39 THEY say first (?)? What was their face doing?**
40
41 >
42
43 **Q6 ? Multiple courts in frame ? does that make video useless today? Their words:**
44
45 >
46
47 **Q7 ? Open to a paid recording session on their court? (yes / maybe / no ? and who
48 to call)**
49
50 >
51
52 ## Required: the most negative thing they said (verbatim)
53
54 >
55
56 ## Best quotes (anything vivid, in their words)
57
58 >
59
60 ## Scores (circle one per row ? anchors in PMF_FIELD_SIGNALS.md $2; when torn, score
lower)
61
62 | Hypothesis | Score |
63 |---|---|
64 | H1 recording behavior | GREEN / AMBER / RED |
65 | H2 dead footage | GREEN / AMBER / RED |
66 | H3 audience | GREEN / AMBER / RED |
67 | H4 whitespace | GREEN / AMBER / RED |
68 | H5 willingness to pay ? their ? number: _____ | GREEN / AMBER / RED |
69 | H6 multi-court pain | GREEN / AMBER / RED |
70 | H7 Phase-2 door | GREEN / AMBER / RED |
71
72 ## Competitor block (only if they named one)
73
74 | | |
75 |---|---|
76 | Product | |
77 | What it does well (their words) | |
78 | What it misses (their words) | |
79 | What they pay | |
80
81 ## Your gut read (circle): COLD · POLITE · CURIOUS · EAGER
82
83 ## Phase-2 candidate? (circle): YES ? call: _____ · MAYBE · NO
84
85 ## Anything else worth knowing (layout, lighting, camera positions you noticed, who
really decides)
86
87 >
88

```

5. user (2026-06-12T14:37:10.584Z)

```

1 # Temporary Engagement: Field Research Associate (Sports-Tech, Bengaluru)
2
3 **Duration:** ~2?3 weeks, part-time (flexible hours, mostly afternoons/evenings when
academies are active)
4 **Location:** Bengaluru ? in-person visits to badminton academies
5 **Compensation:** ? ___ per completed academy visit + travel reimbursement *(discussed

```

individually)*

6 **Reports to:** Avi (phone/WhatsApp: ____)

7

8 ---

9

10 ## The gig in one paragraph

11

12 We're building an AI tool that takes ordinary practice/match video from a badminton academy and automatically finds

13 every rally, builds highlight reels, and generates match reports. Before building further, we need honest, on-the-ground

14 answers from real academies about how they record, what they'd use, and what they'd pay.

15 **Your job: visit 10?15 good

16 academies, have a friendly 15?20 minute conversation with the owner/head coach using the script below, and write down

17 what they actually say.** No selling. No tech knowledge needed. If a conversation goes well, there's a Phase-2 follow-up

18 (coordinating a paid video-recording session) with additional compensation.

19 ---

20

21 ## Phase 1 ? The conversation (what you actually do)

22

23 **Who to talk to:** the academy owner or head coach ? the person who decides what the academy spends money on.

24 If you get a junior coach, be friendly, gather what you can, and ask when the owner is available.

25

26 **Opening (keep it this honest):**

27 > "Hi, I'm helping a Bengaluru engineer who's building an AI tool for badminton academies ? it takes normal practice

28 > video and automatically cuts it into rallies and highlights. He asked me to talk to a few serious academies to

29 > understand how you work with video today. Could I ask you a few questions? Takes 15 minutes, and we're not selling

30 > anything today."

31

32 **The questions (ask in this order; the starred ones matter most):**

33

34 1. Do you ever video-record sessions or matches here? How ? phone, tripod, CCTV, a hired person?

35 2. ? **Do you have recorded footage sitting around that nobody has time to watch?** (multi-court hall recordings,

36 tournament days, CCTV) ? roughly how much?

37 3. Who would watch rally clips or highlights if they existed ? coaches for training? players? parents?

38 4. What do you use today for any of this? (Listen for: Machaxi, VISIST, Game Theory, BadPro+, just YouTube/phone.)

39 What do you like / what's missing in it?

40 5. ? **If you could hand over a full session's video and get back every rally cut out + a highlight reel + a simple

41 report ? would you pay per match for that? What feels fair ? ?100, ?300, ?500 per match?** (Let THEM say a number

42 first. Note their face, not just the words.)

43 6. Your hall has multiple courts running at once, right? Does that make video less useful today?

44 7. Would you let us record one session here (we pay a small fee for the time) to show you what the tool produces on

45 YOUR court? *(This is the Phase-2 door ? just note yes/no/maybe and who to call.)*

46

47 **After each visit, fill ONE answer sheet ? the template is

48 FIELD_VISIT_ANSWER_SHEET.md

49 (print one per visit):** academy name, location, # courts, person spoken to + role + contact, answers 1?7 (their

50 words, not yours), the most negative thing they said (required), exact quotes, your gut

read, and whether they're a
50 Phase-2 candidate. *(How the answers get scored and what totals trigger which decision
lives in
51 [`PMF\_FIELD\_SIGNALS.md`](PMF_FIELD_SIGNALS.md) ? that part is ours, not yours; just
capture honestly.)*
52
53 ---
54
55 ## Ground rules & gotchas (read twice ? this is the difference between useful and
useless)
56
57 1. **You are researching, not selling.** If they ask "how much will it cost", say
"that's exactly what he's trying to
58 figure out ? what would feel fair to you?" Never quote a price yourself.
59 2. **Past behavior beats opinions.** "Would you use video?" gets polite lies. "When did
you LAST record a session, and
60 what happened to that file?" gets the truth. Prefer the second kind of question.
61 3. **Don't promise anything.** No features, no dates, no free trials. The only thing you
may offer is the Phase-2 idea
62 (a paid recording session + free sample output) and only as "would you be open to
it".
63 4. **Write down disappointing answers verbatim.** A "no, we'd never pay for that" with a
reason is worth MORE to us
64 than five vague yeses. You're paid per completed honest visit, not per positive
answer.
65 5. **If they already use a competitor (Machaxi/VISIST/Game Theory etc.) ? goldmine.**
Ask what it does well, what it
66 doesn't, and what they pay. Spend extra time here.
67 6. **Multi-court is our favorite topic.** If they complain that video is useless because
"all the courts are in frame
68 and it's chaos" ? capture that complaint in their exact words. That's the problem we
solve.
69 7. **No recordings of people without clear permission.** In Phase 1 you record NOTHING ?
notes only. Don't photograph
70 the academy or players unless the owner offers.
71 8. **Don't oversit.** 20 minutes, thank them, leave a good impression ? we may come back
to record.
72 9. **Confidentiality:** don't describe the tech beyond the opening line, don't name
other academies' answers, and
73 don't share the answer sheets with anyone except Avi.
74
75 ---
76
77 ## Phase 2 (only for academies that said yes ? separate briefing & pay)
78
79 You coordinate a recording session: we pay the academy a small fee, set up a
phone/camera on a tripod (we provide a
80 one-page capture checklist: behind the court, elevated, whole court in frame, landscape,
1080p/60fps, don't move it),
81 record 122 sessions, and later return a sample highlight reel + report to the academy
for free. **Hard rules:**
82 written consent from the academy AND the adult players being recorded; **sessions with
minors are not recorded for
83 our library** ? adult sessions only; the academy gets told plainly what the footage is
for (building and demonstrating
84 the tool).
85
86 ---
87
88 ## Interested?
89
90 Message Avi at ____ with "FIELD RESEARCH" and your availability. First 2 visits are
paired (you + Avi or a phone
91 walkthrough) so you're never improvising.
92

6. assistant (2026-06-12T14:37:17.816Z)

Now the grounding docs.

7. user (2026-06-12T14:37:19.444Z)

```
34-| | **? PENDING (needs action / not started)** | | | |
35-| C5 | **Output codec royalties** | **P0** | ? | Highlights are still encoded **H.264 + AAC**
('libx264' in `backend\pipeline\stitcher\compiler.py`) ? encode-side AVC patent exposure
remains. | Switch output to **AV1 + Opus** (royalty-free) on an **Ada-class GPU (L4\L40)** for
HW encode. See §4. Decision: provision Ada-class GPU? |
36-| C9 | **Packaging / distribution** | **P0** | ? | No `pyproject.toml`; the diagnostic
`setup.py` shadows the build system. | Design in `DEFERRED_HARDENING_PLAN.md` (#7) ? **awaiting
your approval** before code. |
37-| C12 | **SaaS foundations (async/auth/tenancy)** | **P1** | ? | Sync `/api/process` blocks
on long videos; no auth/tenancy. | Multi-user/platform strategy in `PLATFORM_ARCHITECTURE.md`
(pluggable storage/compute, tenancy, orchestration); async slice = `DEFERRED_HARDENING_PLAN.md`
(#16). |
38-| | **? IN-FLIGHT (in progress / watch)** | | | |
39-| C4 | **Runtime AI strategy (A/B/C + owned model)** | **P0** | ? | Live = **(C) Gemini 2.5
Flash, paid tier**. Long-term = **own a self-hosted generative video-QA model** (fine-tuned open
VLM) as primary, **paid Gemini always-ready as fallback** via the provider registry. | Make
"provider registry + paid fallback" a core principle; build the owned model **after** the
scaffolding/flywheel are solid. See `PLATFORM_ARCHITECTURE.md` §5A. |
40-| C13 | **Beachhead validation** | **P0** | ? | Market research (`PMF_MARKET_RESEARCH.md`)
recommends **India badminton academies** as the #1 beachhead. Unvalidated. | Cheapest tests:
10?15 academy/coach discovery interviews; rally-detection P/R benchmark on 100+ real amateur
clips; fake-door pricing (B2C ~$99/yr, B2B academy ~$600/yr). See §6. |
41-| C8 | **Golden-set vendoring spend** | **P1** | ? | ADR-006: **Roora (Bengaluru)** selected
for an initial paid pilot; licensed/owned video only (no user uploads to vendors). | Confirm
quality-per-dollar from the pilot; then GS-3 `HumanVendorProvider`. See
`GOLDEN_SET_VENDORING.md`. |
42-[Omitted long context line]
43-[Omitted long context line]
44-| C6 | **Inbound HEVC decode** | **P2** | ? ?? | Decoding users' GoPro H.264/**HEVC** uploads
is unavoidable; HEVC pools are fragmented + highest-uncertainty. | Low/uncertain residual;
confirm decode-side exposure with counsel before launch. |
45-| C7 | **Hosting unit-economics** | **P2** | ? | Per-video cost is *cheap* for Gemini (~$0.05
input for a 10-min clip); risk is volume × latency, not unit price. Self-host/offline costs
unmeasured. | **Benchmark $/video & latency on the target VM** for A/B/C before pricing. See §5.
|
46-| C10 | **Value layer (product surface)** | **P2** | ? | Minimal exports shipped
(EDL\JSON\CSV + `basic_stats`, PR #51). | Deepen: per-rally tags/notes that feed golden, per-
player/match stats, clip gallery. See §6. |
--
198-
199---
200-
201-## Changelog
202-
203-- **2026-06-07** ? Added the market-research-of-record (`PMF_MARKET_RESEARCH.md`):
204: ranked beachhead (India badminton academies #1), pricing benchmarks, and a C13
205: "beachhead validation" workstream. Wired the summary into §6.
206-- **2026-06-07** ? Created this living tracker. Merged the 2026-05-30 monetization
207: audit + the commercial-readiness review. **Refreshed since the audit:** C1 AGPL
208: removal now ? shipped (was the #1 blocker); added C5 as a ? (output is still H.264,
209: the audit's AV1 recommendation is not yet implemented); C9 packaging tied to
210: `DEFERRED_HARDENING_PLAN.md`; reflected PR #50/#51 (input-quality reports, value-layer
```

8. user (2026-06-12T14:37:20.524Z)

56:5. **PMF validation (cheaper than any engineering month):** C13 academy interviews (+ the two added questions:

9. user (2026-06-12T14:37:25.567Z)

```
40      Priority order for new videos: **(a) multi-court badminton** (the target distribution
AND where the ship target
41      lives), **(b) ?3 matches per new sport ? table tennis first** (WASB transfers at F1
0.909; pickleball labels are
42      corpus-gold but not trainable until a clean MIT/Apache ball detector exists), (c)
capture variety per
43      `VIDEO_RECORDING_GUIDELINES.md`; **adult sessions only for the training pool**
(consent guardrail). Per video:
44      label ? `curate import-local --normalize --license Proprietary --fps <true fps>` ?
re-run the Gen-0 harness.
45      The paired sign test needs n: at n=10 with 9 wins, p?0.011 ? significance is
reachable this month.
46      2. **[owner decision] Set the ship-gate target** ? now informed: floor = heuristic 0.480
(necessary), **multi-court
47      reference = wrapper 0.66** (the number that makes the owned model worth serving),
clean-footage ceiling = 0.93
48      (not the target ? Gemini serves that tier). Suggested shape: "LOVO F1@tIoU0.5 ? 0.70
on multi-court folds with
49      paired-sign p<0.05 vs heuristic" ? owner to ratify/adjust.
50      3. **M0.4 next increment ($0/local):** swap the Gen-0 LogReg for
**ActionFormer/ASFormer** (MIT, minutes on the
51      2070S ? still lightweight, NOT the heavyweight Gen-2 stack) inside the existing
harness; per-video threshold
52      calibration (the LOVO-vs-oracle gap is visible per fold in the harness output).
53      4. **Multisport thread:** ingest Roora pickleball/TT CSVs (`import-local --normalize`);
merged-prototype LOVO across
54      badminton+TT once TT has ?3 matches; identify a clean pickleball ball detector. ? TT-
on-WASB serving stays gated
55      by C3 (provenance multiplies per sport).
56      5. **PMF validation (cheaper than any engineering month):** C13 academy interviews (+
the two added questions:
57      "multi-court footage nobody watches?" / "pay per-match to index it?"); camera pilot
at the warmest 2-3 academies
58      (consent at capture; demo served via the Gemini path or human-polish ? the reel
pattern exists:
59      `output/MahadevpuraSingles_highlights.mp4`). **Field kit COMPLETE (2026-06-12, owner
ratified the
60      physical-visit approach):** `docs/FIELD_RESEARCH_ASSOCIATE_FLYER.md` (recruit hand-
out: script, ground rules,
61      Phase-2 consent; owner fills ?/contact placeholders) +
`docs/FIELD_VISIT_ANSWER_SHEET.md` (per-visit capture
62      form) + `docs/PMF_FIELD_SIGNALS.md` (G/A/R scoring anchors + **pre-registered
decision rules** ? PROCEED /
63      PIVOT A-C / FALSIFIED thresholds fixed before the first visit).
64      6. **Housekeeping:** ~~rotate the Gemini API key~~ **DONE 2026-06-10** (owner-verified
working; confirm old-key
65      deletion in AI Studio); **rename repo off "badminton"** (owner action,
66      in BACKLOG); collector=SoR confirmation; off-peak niceties: none pending ? the
battery completed.
67
68      ## Pointers / artifacts
69      - Roadmap: `docs/OWNED_MODEL_IMPLEMENTATION_PLAN.md` · Topology: ADR-008
(`docs/archives/decisions/DECISIONS.md`) ·
70      Study: `docs/archives/research/OWNED_MODEL_TRAINING_STUDY.md` · Quality history:
71      `docs/archives/quality-iterations/2026-06-multivideo-lovo/`.
72      - Harness: `training/gen0/harness.py` (one command ? train ? LOVO ? gate ? artifact
bundle; weights gitignored,
73      manifest reviewable). Featurizer: `backend/features/rally_seq.py` (FEATURE_SCHEMA v1).
74      - Corpus: collector DB (system-of-record, 6 matches/262 rallies) · eval manifest
`output/human_lovo_manifest.json`
75      (gitignored) · durable trajectories + Roora files: `C:\Users\avidu\Projects\Annotation
Setup\Collect\Trajectories\` ·
76      labels: `...\Collect\Golden Labelled\`.
```

77 - Gemini ops learnings: serial uploads (`indexing.ai\_max\_workers=1`) + chunk re-encode
78 (`indexing.ai\_chunk\_scale\_width=1280`) + 6-attempt exponential generate retry ? a
cold/prepaid project burst-throttles
79 with a MISLEADING "credits depleted" 429; ramp gradually.

10. user (2026-06-12T14:37:27.195Z)

docs\FIELD_VISIT_ANSWER_SHEET.md:81:## Your gut read (circle): COLD · POLITE · CURIOUS · EAGER
docs\FIELD_RESEARCH_ASSOCIATE_FLYER.md:59:2. **Past behavior beats opinions.** "Would you use
video?" gets polite lies. "When did you LAST record a session, and
docs\PMF_FIELD_SIGNALS.md:28:| H7 | **Behavioral commitment** ? they'll let us record on their
court (Phase-2 door) | polite no across the board | Q7 |
docs\PMF_FIELD_SIGNALS.md:53:polite conversation, not a customer.
docs\PLATFORM_ARCHITECTURE.md:255: **Bias: start with the simplest thing that gives retries +
status; graduate only on
docs\PLATFORM_ARCHITECTURE.md:533:4. **Orchestration tech** (see research recommendation) ? bias
to the simplest that gives
docs\UI_PROPOSAL.md:14:multi-court-biased footage ? the distribution where the owned model's
0.66 ship target lives.

11. user (2026-06-12T14:37:32.451Z)

19:## 1. Executive summary
57:## 2. Market sizing & trends
59:### 2.1 Sports analytics / sports-video market
71:### 2.2 Action-camera / amateur recording adoption
80:### 2.3 Participation & popularity by sport and geography
111:## 3. Customer segment evaluation
130:## 4. Competitive & pricing benchmarks
155:## 5. Recommended beachhead
160:### #1 (strongest) ? Badminton coaching academies & serious players, **India**
168:### #2 ? Serious pickleball players & clubs, **US (then UK/Australia/Asia)**
174:### #3 ? Badminton academies/clubs, **East/SE Asia (China, Indonesia, Malaysia, Japan,
Korea) + Denmark**
187:## 6. Risks & open questions (what would invalidate the thesis)
215:## 7. Sources

12. user (2026-06-12T14:37:39.799Z)

130 ## 4. Competitive & pricing benchmarks
131
132 | Product | Category | Model | Price | Notes (year) |
133 |---|---|---|---|---|
134 | **Veo (Cam 3)** | Team-sports auto-filming | Hardware + sub | Camera ~USD 1,533 (5G
~1,886) + ~£395?895/yr | 15,000+ clubs/80 countries; \$115M raised (2025) |
135 | **Trace** | Soccer auto-filming | **Free camera + sub** | USD 180/yr (Basic) ? 300
(Pro) | Camera leased free with sub (2025) |
136 | **Pixellot (Air)** | Auto camera/streaming | Hardware + sub | Camera ~USD 949 + USD
69?167/mo | Institutional/league; 10,000+ clubs (2025) |
137 | **Spiideo** | Auto-filming/analytics | Sub incl. camera | ~USD 3,800/yr (1 team) ?
6,100/yr (club) | Institutional budgets (2025) |
138 | **Hudl** | Team video analysis | SaaS tiers | From ~USD 400/yr | Broad team-sports
incumbent (2026) |
139 | **SwingVision** | **Tennis/pickleball AI, single iPhone** | Software freemium | **USD
179.99/yr**; free 2 hrs/mo | Closest analog; 200k+ monthly users; \$10M+ raised (2025) |
140 | **BadPro+** | **Badminton AI on phone** | **Pay-per-use, no sub** | Per-analysis (? "<
one coaching session") | Direct badminton competitor; shot detection, heatmaps, auto-highlights
(2025) |
141 | Niche pickleball/badminton apps | Various | Mixed | Varies | Fragmented; few true
auto-highlight tools (2025) |
142
143 **What buyers actually pay (synthesis):** consumer software-only anchors at ~USD
144 100?200/yr (SwingVision 179.99); hardware-bundled USD 180?900/yr prosumer/club,
145 USD 3,800?6,100/yr institutional; badminton-specific BadPro+ uses *pay-per-use* ?

signaling a real badminton AI market but ****unproven subscription WTP**** (opportunity + warning). ****Positioning:**** software-only + BYO-camera undercuts hardware players and sells worldwide via SaaS. Realistic anchors ****(estimate):** B2C USD 60-180/yr; B2B academy/club USD 300-1,500/yr** per location. The differentiators (cheap offline self-improving detection, per-rally stats/tagging, clean licensing) support a ****B2B**** value story more than a B2C one.

5. Recommended beachhead

Ranked by ****WTP × participation density × existing recording behavior × static-camera fit × competitive whitespace.****

#1 (strongest) ? Badminton coaching academies & serious players, **India**
 20M+ recreational players (2023); rapidly ****formalizing academy ecosystem**** + Khelo India tailwind; academies already record fixed-angle; coaches are buyers who monetize instruction (high relative WTP); incumbents ignore badminton; the only direct badminton AI (BadPro+) is early with unproven pay-per-use. Product is ****native badminton-first****. ****Watch-outs:**** price sensitivity (need INR-tuned tiers); sell to the ****academy****, not the individual.

#2 ? Serious pickleball players & clubs, **US (then UK/Australia/Asia)**
 Fastest-growing sport (19.8M US, +45.8% YoY, 2024); capital-rich; ~70,000 courts; strong recording/sharing culture; pickleball is on the roadmap. ****Watch-out:**** ****SwingVision** already owns single-camera tennis+pickleball AI** at USD 179.99/yr (200k+ monthly users) ? head-to-head differentiation, not greenfield.

#3 ? Badminton academies/clubs, **East/SE Asia (China, Indonesia, Malaysia, Japan, Korea) + Denmark**
 Highest global density (China 80M+; Denmark 100k+ structured club members; Japan/Korea school-club & workplace systems), mature club infra, fixed-court recording. ****Watch-outs:**** China needs local hosting/partners + is regulatory-complex; SE Asia price-sensitive; Denmark/EU small but high-WTP and easy to pilot.

****Bottom line:**** lead with ****India badminton academies**** (whitespace + behavior + buyer alignment), use ****enthusiast B2C as the funnel****, validate a ****club/academy B2B2C motion****, then expand to ****pickleball-US**** (out-build SwingVision) and ****badminton-Asia/Denmark**** (density).

6. Risks & open questions (what would invalidate the thesis)

- **Tripod/static-camera adoption among amateurs.**** If serious amateurs won't set up a fixed camera, B2C TAM collapses to coaches/clubs. ****Validate:**** observe 3-5 academies + 50-100 enthusiasts on whether they already record fixed-angle.
- **Free / near-free alternatives**** (YouTube, free app tiers, BadPro+ pay-per-use) cap B2C WTP. ****Validate:**** fake-door pricing at USD 60-180/yr.
- **Capture-quality variance = the failure mode**** (gym lighting, off-center angle, occlusion, mixed resolutions). ****Validate:**** benchmark rally-detection P/R on 100+ real amateur clips; define a "good-capture" spec + auto-reject (ties to `VIDEO_RECORDING_GUIDELINES.md` + the observability reports).
- **Monetization mismatch in best-participation markets**** (India/Indonesia/China are price-sensitive, hard to bill). ****Validate:**** local pricing/payments pilot; per-academy INR licensing.
- **Competitive response**** (SwingVision or a team-sports incumbent adds racket sports). ****Validate:**** ship a badminton-quality advantage incumbents can't trivially match (offline self-improving detection, per-rally tagging).
- **B2B sales-cycle drag**** for clubs/federations/schools. ****Validate:**** single-coach/academy self-serve before institutional deals.
- **Clean-licensing positioning**** matters mainly to B2B buyers ? confirm it's a real purchasing criterion via 5-10 buyer interviews.

****Cheapest next steps:**** (a) 10-15 discovery interviews with Indian badminton academy

210 owners/coaches; (b) a capture-quality benchmark on real amateur footage; (c) a fake-door
211 pricing test across B2C (~USD 99/yr) and B2B academy (~USD 600/yr).
212
213 ---
214
215 ## 7. Sources
216
217 - Sports analytics market: Precedence Research; Fortune Business Insights; Grand View
218 Research; MarketsandMarkets; Future Market Insights. *(2024?2025)*
219 - <https://www.precedenceresearch.com/sports-analytics-market>

13. user (2026-06-12T14:39:22.466Z)

Structured output provided successfully

14. assistant (2026-06-12T14:39:30.407Z)

Verdict returned: the claim is real at medium severity ? the flyer's opening does reveal the solution before the past-behavior questions, no pre/post-reveal tracking exists anywhere in the kit, and two of three PROCEED legs are courtesy-inflatable at these count thresholds, though the behavioral H7 anchor and downstream Phase-2 gate bound the damage.